Weighted Range-Constrained Ising-Model Decoder for Quantum Error Correction

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***Abstract*—Ising model-based Quantum Error Correction decoders reduce topological complexity compared to classical decoders. However, the SOTA Ising decoder has a higher time complexity than union-find (UF) and a lower threshold than minimum-weight perfect-matching (MWPM). We propose the Weighted Range-Constrained Ising Model-Based (WRIM) decoder. WRIM uses a polygonal region to enclose flipped syndromes, ensuring the coverage of all potential error chains while optimizing coupling and external field coefficients. WRIM reduces the variable count by 97.8x, achieves microsecond-level decoding, and has a worst-case time complexity of $O(n)$, outperforming UF. WRIM exhibits threshold behavior up to 10.7-11.0%, surpassing the MWPM's highest reported threshold.**

***Index Terms*—Quantum error correction, Ising model, fast decoding method, surface code, variable reduction.**

I. Introduction

Quantum computers introduce a new computing paradigm with the potential to solve problems that are intractable in polynomial time on classical systems [1], [2]. One of the main barriers to realize practical quantum computers is decoherence—a loss of quantum coherence that occurs when qubits interact with their environment [3], [4]. Therefore, achieving reliable quantum computation requires fault-tolerant quantum computing (FTQC), with quantum error correction (QEC) as its cornerstone to detect and correct errors during computations [5], [6].

In QEC, multiple physical qubits are combined to form a single logical qubit, thereby enhancing resistance to quantum errors. This method of constructing logical qubits is called a "code," and various methods have been proposed. Among them, Kitaev's surface code [7] and its variants [8], [9] are considered the leading candidates for realizing QEC due to their superior quantum error resilience and compatibility with integrated qubits, as they only require gate operations with neighboring qubits. The surface code utilizes two types of qubits: data qubits for information storage and ancilla qubits for error detection.

In the surface code, "decoding" is necessary to estimate the data qubits where errors have occurred based on the measurements of the ancilla qubits (referred to as syndromes). Effective QEC decoders must operate fast enough to prevent error accumulation [10] (e.g., on a microsecond timescale for superconducting transmon qubits [11], [12] and on a millisecond timescale for trapped ions [13]) and achieve target logical error rates.

Numerous QEC decoders have been developed, such as Minimum-Weight Perfect Matching (MWPM) decoders [14]–[16], which transform the decoding problem into a graph matching problem. However, MWPM decoders have a worst-case time complexity between $O(n^3)$ and $O(n^7)$ relative to the number of data qubits n, limiting the code distances d at most 9 when adhering to real-time latency constraints [15], and up to 33 with fast MWPM when assuming a circuit-level noise of 0.1% [17]. Additionally, these decoders demand substantial memory resources [18]. The tensor network decoder [19] reformulates the decoding problem as a machine learning problem and solves the decoding problem with a time complexity of $O(n^2)$. Neural network (NN) decoders [20]–[22] also show promise, though training a suitable NN for QEC demands substantial computational resources and time. Currently, the fastest decoder is the Union-Find (UF) decoder [23], [24], which leverages the union-find data structure and has a worst-case runtime of $O(n\alpha(n))$ ($\alpha(n) \leq 3$). However, when implemented on Systolic Arrays, it is constrained to code distances due to limited memory and processing capacity, handling up to $d = 15$ [25].

Recently, an Ising model-based decoder was introduced to address these challenges [26], [27], framing QEC decoding as an Ising model task solvable by annealing machines [28], with scalability up to a code distance of 46. However, despite its polynomial computational scaling slightly better than MWPM decoders, its decoding time remains suboptimal.

This paper introduces the Weighted Range-Constrained Ising Model-Based (WRIM) decoder, a fast Ising model-based approach for efficient QEC. The WRIM decoder employs a polygonal region to enclose flipped syndromes with their neighbors, aiming to limit the number of nodes while ensuring the connectivity of all flipped syndromes. By limiting qubits in this region, the Ising model for error decoding is formulated to capture all potential error chains and possible error sites. It then assigns optimal weights to the Ising model's coupling coefficient and external field coefficient based on the satisfaction of syndrome parity for each error pattern, aiming to minimize the length of each error chain.

This paper presents the following novel contributions.

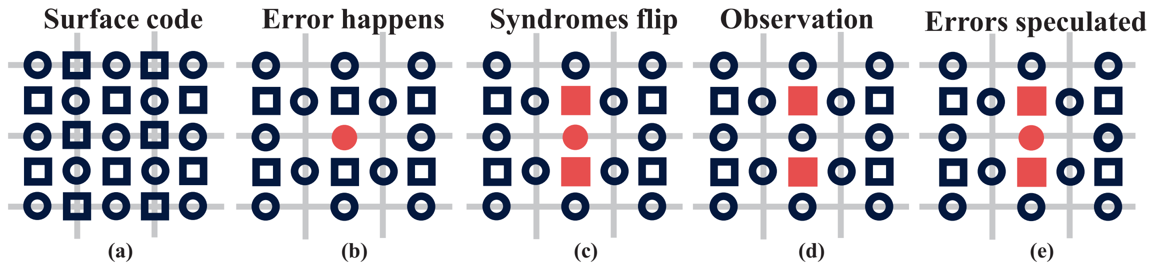

Fig. 1. (a) Planar surface code with code distance $d = 3$. Along the gray lines representing the surface code, open circles indicate data qubits, while squares denote X and Z ancilla qubits. X ancilla qubits are positioned on the lattice grid, while Z ancilla qubits are positioned between the grids. (b)–(e) depict the cases with only X errors. An X error on a data qubit is represented as a red-filled circle (b), flipping the syndrome from $+1$ to -1, thereby turning the open blue squares into filled red squares (c). By analyzing these Z syndromes measured by the Z ancilla qubits (d), the location of X error can be identified (e).

- The WRIM decoder achieves up to a 97.8x reduction in the number of variables compared to the state-of-the-art (SOTA) Ising model-based decoder [26], significantly decreasing computational resource demands.
- The WRIM decoder's region formation has a time complexity of $O(kn)$ with $k \leq 1$, and the worst-case complexity is $O(n)$. The WRIM decoder's Ising decoder has a time complexity of $O(n^a)$, with $a \leq 1$. Specifically, at a 20% physical error rate, its complexity is $O(n^{0.97})$, which represents the worst-case scenario in practice. This results in an overall $O(n)$ complexity in WRIM, which is lower than $O(n \log n)$ complexity of the fastest SOTA decoder, UF with a worst-case complexity of $O(n\alpha(n))$ where $\alpha(n) \leq 3$. Experiments on existing quantum annealers [29] at 0.1% and 1% physical error rates demonstrated decoding time in the order of microseconds.
- By retaining nearly all error chains and adaptively assigning Ising-weights for each error pattern, WRIM achieves a logical error threshold of 10.7-11.0%, exceeding the highest threshold of 10.3% achieved by the MWPM decoder, demonstrating better fault tolerance.

In addition to these contributions, it is important to note that quantum decoding, particularly in QEC, poses significant circuit design challenges. Addressing these challenges via techniques in electric design automation is crucial for advancing quantum computing and unlocking its potential to surpass classical computation.

The rest of the paper is organized as follows: Sec. II provides a brief introduction to surface codes and the mapping of the QEC decoding to the Ising model. Sec. III details the proposed WRIM decoder. Sec. IV presents the experimental results, including the number of variables needed for WRIM decoder, decoding time, and logical error rate threshold. Sec. V concludes the paper.

II. Surface Code and Ising Model

A. Surface Code and Quantum Error Correction

This paper focuses on the planar surface code [7], [8], widely regarded for its practical implementation potential. However, the proposed decoder is also applicable to other surface codes, such as the rotated surface code [30], [31] and the color code [32].

In QEC, as depicted in Fig. 1 (a), the linear length of the lattice defines the code distance. The lattice comprises two qubit types: data qubits (circles on lattice edges), and ancilla qubits (squares). These qubits are prone to errors induced by various noise sources, which can result in bit-flip errors (X errors) or phase-flip errors (Z errors). Ancilla qubits are placed to detect errors in data qubits without directly observing data qubits. There are two types of ancilla qubits: X ancilla qubits, located on the lattice grid, detect Z errors, while Z ancilla qubits, positioned between grids of the lattice, are designed to detect X errors.

Syndromes represent the collective logical state of data qubits as measured by ancilla qubits. Syndrome $+1$ indicates there is no error or an even number of errors, while -1 reflects an odd number of errors in the surrounding data qubits. As shown in Fig. 1 (b)–(e), an X error on a data qubit changes its representation from an open circle (blue) to a filled circle (red). This flips syndrome from $+1$ to -1, altering the corresponding ancilla qubit's symbol changes from an open square (blue) to a filled square (red). The QEC decoder deduces error locations from these syndrome observations.

For a planar surface code with code distance d, the number of data qubits is $d^2 + (d-1)^2$, while the number of X and Z ancilla qubits is both $d \times (d+1)$. This arrangement ensures a Hilbert space dimension of 2, which represents a logical qubit [33]. Therefore, the Z ancilla qubits in the top and bottom rows, as well as X ancilla qubits in the left and right columns, are each connected to three data qubits.

For simplicity, X and Z errors are assumed to occur independently with probability p, allowing them to be corrected separately. In this paper, we focus solely on X errors detected by Z ancilla qubits. The treatment of Z errors is entirely analogous, as rotating the surface code by 90 degrees swaps the positions of X ancilla qubits with those of Z ancilla qubits.

B. Error Pattern

An error in a data qubit flips two adjacent syndromes, as illustrated in Fig. 2 (a). If a connected syndrome does not flip (Fig. 2 (b)), it indicates errors in an even-numbered of

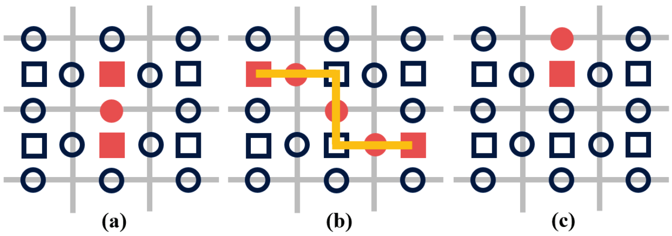

Fig. 2. Error patterns. (a) An error can flip two connected syndromes. (b) Neighboring errors generate an error chain, and the syndrome flips at both ends of the error chain. (c) The error in the first and last rows of the surface code will only flip one syndrome.

connected data qubits, forming an error chain. These chains cause the syndromes at their endpoints to flip. Any error chain connecting the flipped syndromes (without crossing the square) is a valid solution. A special case, known as a boundary condition, occurs when errors in data qubits on the first or last row flip only one syndrome (Fig. 2 (c)). In such cases, syndromes can form chains with other syndromes or the boundary.

C. Ising Model for Quantum Error Correction

The Ising model-based decoder [26] transforms the QEC into the following Ising model:

$$H = -J \sum_{v}^{N_v} b_v \prod_{i \in \delta v}^{4 \text{ or } 3} \sigma_i - h \sum_{i}^{N_d} \sigma_i .$$

Data qubits are represented by variables σ_i, which flip from $+1$ to -1 during the annealing process if an error occurs. An ancilla qubit is denoted by v, with δv representing the list of connected qubits. The syndrome b_v is $+1$ for even or no errors and -1 for odd errors. N_v and N_d represent the numbers of syndromes and data qubits, respectively.

The first term, with coupling coefficient J, ensures that the decoded error pattern aligns with the syndromes for accurate correction. The second term, with external field coefficient h, minimizes the total errors by favoring solutions with fewer flips. Setting $J > h$ guarantees the correctness of the solution. Thus, the lowest energy state of the Ising model represents an error pattern that fits the syndromes while minimizing the total number of errors.

Algorithm 1: Range-Constrained Polygon Region Formation

Input: Flipped syndrome set S_F

Output: Dictionaries Boundary and Row recording the region.

```
1  Create three dictionaries Row, Col and Boundary and
     one set Non_empty.
2  foreach S_F do
3      Update Row[row_i] and Col[col_j] ;
4      Row[row_i] = [min(col_i), max(col_i)] ;
5      Non_empty add row_i ;
6      Col[col_j] = [min(row_j), max(row_j)] ;
7      if min(row_j) ≠ max(row_j) then
8          Boundary[col_j] = [0, 2d − 1] ;
9      else
10         if min(row_j) + max(row_j) < 2d − 1 then
11             Boundary[col_j] = [0, min(row_j)] ;
12         else
13             Boundary[col_j] = [max(row_j), 2d − 1] ;
14 for row ∈ [min Row.keys(), max Row.keys()] do
15     if row ∉ Non_empty then
16         Find nearest upper row up_row
17         and lower row lo_row in Non_empty ;
18         min_inter = max(Row[up_row][0],
             Row[lo_row][0]) ;
19         max_inter = min(Row[up_row][1],
             Row[lo_row][1]) ;
20         if max_inter - min_inter ≥ 2 then
21             Row[row] = [min_inter, max_inter] ;
22         else
23             min_union = min(Row[up_row][0],
                 Row[lo_row][0]) ;
24             max_union = max(Row[up_row][1],
                 Row[lo_row][1]) ;
25             Row[row] = [min_union, max_union] ;
26             Row[up_row] = [min_union, max_union] ;
27             Row[lo_row] = [min_union, max_union] ;
```

III. Weighted Range-Constrained Ising Model-Based Decoder

Unlike the Ising model-based decoder [26], which includes all syndromes and qubits across the entire surface code, the WRIM decoder defines a polygonal region around flipped syndromes. This region ensures the shortest error chains between flipped syndromes and encompasses potential error chains meeting boundary conditions, as shown in Fig. 2 (c).

The WRIM decoder operates in three stages. First, it determines allowable row and column ranges for flipped syndromes in the surface code. Second, it constructs an Ising model using syndromes and variables within this region. Finally, multiple weights for coupling and external field coefficients are applied to the annealing machine. The solution with the largest external field coefficient satisfying syndrome parity identifies the most likely error locations.

A. Range-Constrained Polygon Region Formation

The process of forming the range-constrained region is detailed in Algorithm 1 and illustrated in Fig 3. In steps 2–6, each flipped syndrome updates row and column dictionaries to define the range of columns for each row and rows for each column, encompassing all flipped syndromes. Steps 7–13 handle boundary conditions: if a column includes a single flipped syndrome, the closer boundary is connected; if it includes two or more, both upper and lower boundaries are linked. Steps 14–27 ensure row connectivity by linking flipped syndromes across rows. If two rows have at least two columns, the path

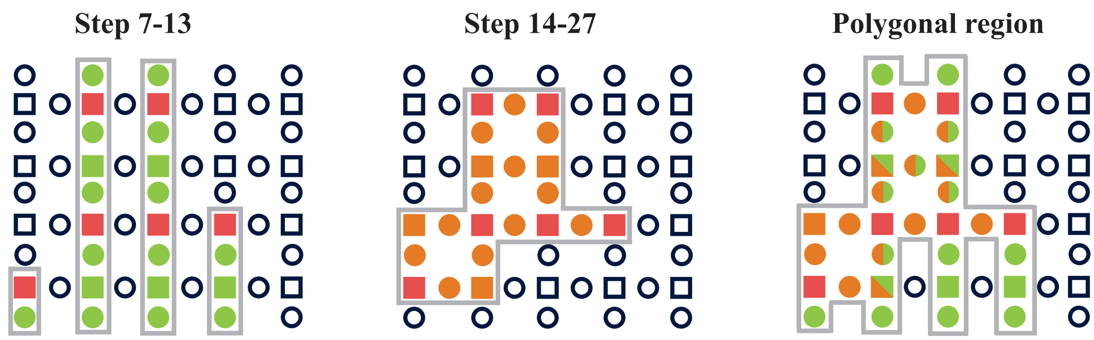

Fig. 3. Steps 7-13 include the non-flipped syndromes and qubits located between flipped syndromes and boundaries, represented as green-filled squares and circles. Steps 14–27 add non-flipped syndromes and qubits forming paths between flipped syndromes across rows, represented as orange-filled squares and circles. In the polygonal region, squares and circles in both colors indicate areas where non-flipped syndromes and qubits are included during both steps.

TABLE I
THE WEIGHT SETTING WITH DIFFERENT LENGTH OF ERROR CHAIN (J = 1024)

Length	1–2	3–4	5–6	7–8	>8
h	900	400	260	20	1

width is set to the intersection; otherwise, the row ranges are merged to determine the maximum range, maintaining nearly all paths.

The resulting polygonal region includes the error chains between flipped syndromes and satisfies boundary conditions, ensuring it captures the necessary and sufficient information to locate errors accurately.

B. Range-Constrained Ising Model-Based (RIM) Decoder

After determining the polygonal region, both syndromes and qubits within the regions are configured to form the Ising model. All syndromes are represented as b_v, while the connected data qubits are designated as σ. The Ising model is constructed as:

$$H = -J \sum_{v \in \text{region}} b_v \prod_{i \in \delta v}^{1\sim 4} \sigma_i - h \sum_{i \in \text{region}} \sigma_i$$

The value of J must ensure that the error pattern matches the expected error syndrome, while h aims to reduce the number of errors.

In this RIM decoder, not only are syndromes and qubits outside the region excluded, but the Ising model no longer requires new variables to reduce the order of products involving four qubit variables. For an Ising machine, handling terms beyond quadratic is impractical, necessitating the introduction of new variables to reduce higher-order terms. For example, a cubic term $x_1x_2x_3$ is transformed into quadratic terms using a penalty function: $x_1x_2x_3 = yx_3 + P \cdot (x_1x_2 - 2x_1y - 2x_2y + 3y)$ [34]. Here, y is the newly introduced variable and P is the penalty value that enforces $y = x_1x_2$.

C. Weighted RIM (WRIM) Decoder

For an error chain of length l, the change in Hamiltonian energy caused by flipping the corresponding variable must be negative, i.e., $-2J + lh<0$ [26]. By appropriately setting h and J, the Hamiltonian energy landscape can be optimized for efficient solution finding. For error chains of length less than or equal to 2, J is set slightly greater than h, creating a distinct energy gap between global and local minima. For the longest error chain of length 3–4, J should be adjusted to slightly exceed $2h$. Table I provides parameter settings of h for various chain lengths with J fixed at 1024.

If the longest error chain in the decoded error pattern exceeds the length used for weight settings (J and h coefficients), the annealing solution may fail to satisfy syndrome parity. To address this, the Weighted RIM (WRIM) decoder applies the Ising model simultaneously across all weight settings in Table I. Models with larger values of h solve faster. If a valid solution satisfying syndrome parity is found under such a model, it is accepted, and further searches with different weights are halted. Otherwise, the process continues until a suitable solution is identified.

IV. Numerical Experiments

The experiments were conducted using Python 3.10.12 on a classical computer with a Core i9-13900KF CPU and 125 GB of memory for Ising model construction. The model was solved using two methods: simulated quantum annealing (SQA) using a GPU annealing cloud server—Amplify Annealing Engine (AE) [35] and real quantum annealing (QA) via the D-Wave cloud server [29]. In AE, the annealing time determines the duration of each process, with numerous annealing iterations performed to locate the lowest energy state. Increased problem complexity and variable count increases result in longer iteration times. After annealing on the cloud server, solutions are returned to the classical computer. D-Wave, employing superconducting qubits, allows fine control over annealing durations and iteration counts. The Ising model is sent to the D-Wave cloud, where QA is performed to find the lowest energy state, with solutions subsequently returned to the classical computer.

TABLE II
THE COMPARISON OF NUMBER OF VARIABLES BETWEEN [26] AND WRIM

d	qubits	[26]	WRIM with physical error rate (%) 1	2	5	10	15
5	41	109	2.35	5.63	20.94	41.82	48.73
7	85	247	9.18	21.9	79.58	112.48	135.18
9	145	441	31.73	70.99	159.96	217.17	254.08
11	221	691	74	154.6	253.01	353.43	444.56

TABLE III
TIME COMPLEXITY FOR THE FIXED PHYSICAL ERROR RATES OF THE WRIM, ISING [26], AND MWPM DECODERS.

p (%)	WRIM's Ising	WRIM	Ising [26]	MWPM	UF
0.1	$O(n^{0.002})$	$O(n^{0.002}) + O(kn)$	$O(n^{1.01})$	$O(n^{2.06})$	$O(n \log n)$
1.0	$O(n^{0.01})$	$O(n^{0.01}) + O(kn)$	$O(n^{1.79})$	$O(n^{2.10})$	
2.0	$O(n^{0.04})$	$O(n^{0.04}) + O(kn)$	$O(n^{1.74})$	$O(n^{2.24})$	
5.0	$O(n^{0.09})$	$O(n^{0.09}) + O(kn)$	$O(n^{1.84})$	$O(n^{2.50})$	
10	$O(n^{0.24})$	$O(n^{0.24}) + O(kn)$	$O(n^{1.81})$	$O(n^{2.72})$	
20	$O(n^{0.97})$	$O(n^{0.97}) + O(kn)$	$O(n^{1.54})$	$O(n^{2.83})$	

A. Number of variables

The number of variables in WRIM was compared to the Ising model-based decoder [26] across code distances ranging from 5 to 11 using 100 random error patterns per distance, and physical error rates of 1%, 2%, 5%, 10%, and 15%. In [26], the variable count remains constant for a given code distance, as all syndromes and data qubits are considered. In contrast, WRIM's variable count varies with error patterns and physical error rates. The average number of variables for each distance and error rate was recorded. Table II details the code distance (d), the number of data qubits (qubits), the variable count for the Ising model-based decoder [26], and WRIM's average variable count under different error rates.

As the code distance increases, so does the number of variables. The WRIM decoder significantly reduces the required variables compared to [26], with fewer variables needed at lower physical error rates. This reduction occurs because WRIM excludes syndromes and qubits outside the defined region and simplifies the Ising model by eliminating the need for additional variables to reduce higher-order terms.

B. Decoding Time

Decoding time is critical as qubit states may collapse before decoding completes. The decoding time of the WRIM decoder consists of two components: the time to form range-constrained regions and the time required by the annealing machine to solve the Ising model. Communication time between classical computers and cloud servers, such as AE and D-Wave, are excluded, as the QEC decoders as well as qubit controllers are expected to be deployed near qubits to enable high-fidelity and real-time error correction [13], [36].

1) Range-Constrained Region Formation Time:

Algorithm 1 shows a worst-case time complexity of $O(n)$. Experiments were conducted for d values from 3 to 12 and physical error rates of 1%, 5%, 10%, 15%, and 20%. The

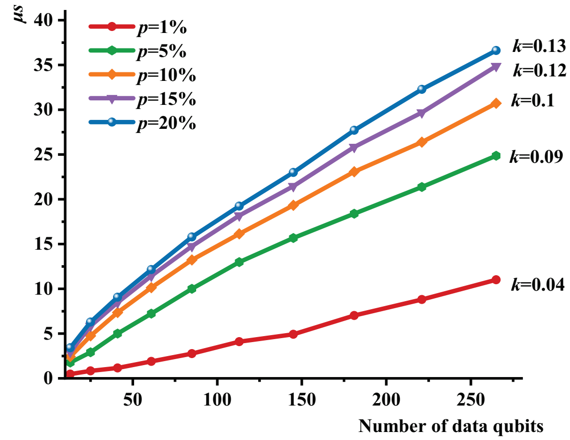

Fig. 4. Region formation time as a function of data qubits count under different physical error rates.

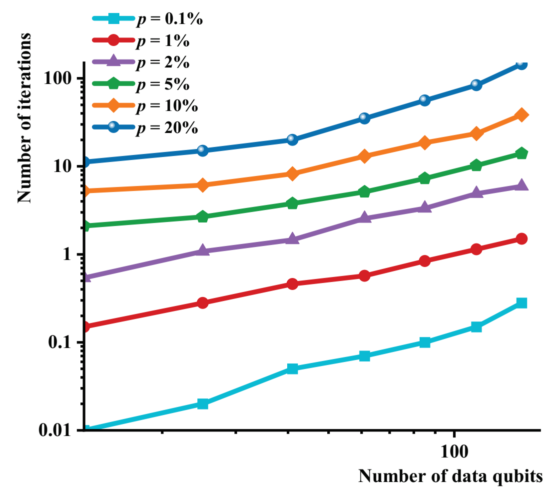

Fig. 5. Number of iterations as a function of data qubits count under different physical error rates.

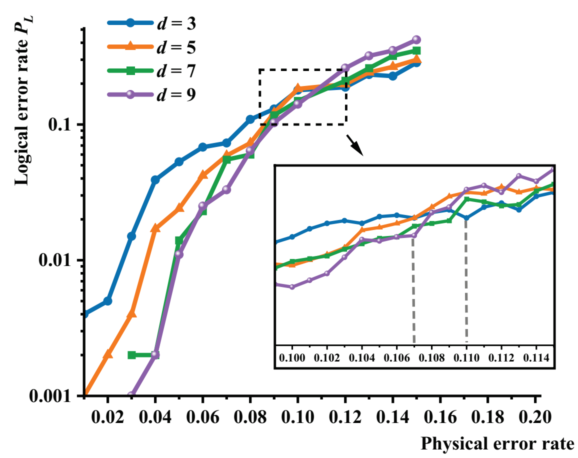

Fig. 6. Correlation between the calculated logical error rate P_L and the physical error rate p for d from 3 to 9.

TABLE IV
THE DECODING TIME USING QUANTUM ANNEALER D-WAVE.

d	p (%)	link time (μs)	annealing time (μs)	decoding time (μs)
5	0.1	0.26	0.03	0.29
7	0.1	0.33	0.05	0.38
9	0.1	0.89	0.10	0.99
5	1	7.84	37.90	45.74
7	1	13.28	63.50	76.78
9	1	14.46	195.30	209.76

time required to form regions is shown in Fig. 4. The time complexity with respect to the number of data qubits is $O(kn)$, with k increasing as the physical error rate rises.

2) Decoding Time by Ising Machine:

The average number of iterations for d from 3 to 9 with physical error rates of 0.1%, 1%, 2%, 5%, 10%, and 20% were tested. The results are presented in the double logarithmic plot in Fig. 5. The plot exhibits approximately linear behavior, indicating that the number of iterations scales as a polynomial function of the number of data qubits. Through regression analysis, the polynomial degree is found to be 0.002 (minimum) for $p = 0.1\%$ and 0.97 (maximum) for $p = 20\%$.

We examine the time complexity of the WRIM decoder, including range-constrained region formation and Ising annealing iterations, and compare it with other decoders, including the Ising model-based decoder [26], MWPM [26] and UF, as shown in Table III. Compared to SOTA Ising-model decoder [26], the reduction in time complexity of WRIM results from eliminating unnecessary variables, lowering the overall problem complexity significantly. Additionally, the weight (h and J) settings also help distinguish between the global minimum and local minima during the solution search, further reducing the search complexity. As a result, the complexity of decoding the Ising model is reduced. Compared to currently the fastest decoder UF, it is evident that the time complexity of iteration in WRIM is less than $O(n)$, resulting in an overall time complexity $O(n)$, which is lower than the time complexity of UF ($n \log n$).

The actual decoding time is also crucial for practical applications. The machines that solve the Ising model such as the SQA machine AE [35], simulated annealing like OpenJij [37], or true QA machines like D-Wave [29], have varying annealing times. In this experiment, real QA machine D-Wave is used to test the average annealing time for 100 error patterns at physical error rates of 0.1% and 1% to demonstrating the potential of QA and the WRIM decoder, as shown in Table IV.

At physical error rates of 0.1% and 1%, consistent with current advanced error rates ranging from 0.1% [11] to 1% [38], WRIM achieves decoding speeds on the order of microseconds, meeting the high-speed requirements for superconducting transmon qubits. It is important to note that QA machines are still in the developmental stage, particularly in terms of hardware scale and graph topology. As QA technology advances, the decoding speed of WRIM is expected to improve, along with an expansion in the decoding scale.

C. Logical Error Rate

The logical error rate is another key metric for QEC decoders. We evaluate the logical error rate threshold across code distances from 3 to 9. The threshold is defined as the physical error rate below which the logical error rate can be reduced by increasing the code distance. For each physical error rate, 1000 samples were tested to calculate the logical error rate.

The results are shown in Fig. 6. The threshold for the WRIM decoder lies between 10.7% and 11.0%, which is higher than the highest threshold of 10.3% achieved by the MWPM. The reason is that, under the condition of satisfying syndrome parity, weights with larger values of h tend to find error patterns that shorten each individual error chain, rather than minimizing the total number of errors. In contrast, the MWPM minimizes the total weight of the error chain, which typically results in patterns that reduce the overall error count but may not be the shortest for each error chain.

V. CONCLUSION

This paper proposed the Weighted Range-Constrained Ising Model-Based Decoder (WRIM), a fast Ising model-based decoder for decoding surface codes. The WRIM decoder employs a polygonal region to enclose flipped syndromes with their neighbors, limiting the number of nodes while ensuring the connectivity of all flipped syndromes. By constraining qubits to this region, the Ising model for error decoding is formulated to capture all potential error chains and error sites. It then assigns optimal weights to the Ising model's coupling coefficients and external field coefficient based on the satisfaction of syndrome parity for each error pattern, aiming to minimize the length of each error chain. Experimental results demonstrate the superior performance of WRIM, achieving up to a 97.8x reduction in variables compared to SOTA Ising model-based decoders while also enabling microsecond-level decoding at 0.1% and 1% physical error rates on the real quantum annealer D-Wave. Furthermore, the region algorithm of WRIM on a classical computer has a complexity of $O(kn)$ ($k \leq 1$), with decoding by Ising machine achieving $O(n^a)$ ($a \leq 1$), resulting in overall time complexity of decoding time $O(n)$. This is lower than the $O(n \log n)$ time complexity of the current fastest decoder, UF. Additionally, because WRIM uses weight to ensure that every error pattern has relatively shorter error chains rather than globally shortest chains, WRIM exhibits threshold behavior, maintaining performance at physical error rates up to 10.7-11.0%, making it surpass the highest threshold decoder MWPM.

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